FitBuddy-AI Fitness Plan Generator using Gemini Models

Team ID: UNM-293 2026-2430

Team Member: Kaviya sree.T

Project Description

FitBuddy is a Generative AI powered fitness planning assistant designed to provide personalized fitness plans based on user-provided goals, preferences, fitness level, available time, and other relevant inputs. The application uses Gemini Models to generate structured and easy-to-follow fitness guidance through a simple user interface.

The system is intended to organize workout suggestions, basic activity guidance, and personalized planning into one convenient application. The generated plan can be adapted to the information supplied by the user.

Scenario

A user enters their fitness goal, current fitness level, preferred activities, available time, and other relevant details. FitBuddy processes these inputs through the Gemini model and generates a personalized fitness plan that is presented clearly on the application.

Technical Architecture

The application follows a modular architecture consisting of a frontend, backend, and Gemini AI integration. The frontend collects user inputs and displays the generated plan. The backend validates the input, constructs a structured prompt, communicates with the Gemini API, and returns the generated result to the frontend.

Suggested components: HTML/CSS/JavaScript frontend, Python Flask backend, Gemini API integration, secure environment-variable based API-key management, and optional database support if the project stores user plans or history.

Pre-requisites

• Python programming fundamentals

• Flask framework knowledge

• HTML, CSS, and JavaScript basics

• Gemini API familiarity

• Prompt engineering basics

• Git/version-control basics

• Development environment setup

Project Workflow

Milestone 1: Model Selection and Architecture

• Activity 1.1: Create and securely configure the Gemini API key.

• Activity 1.2: Research Gemini models and select a suitable model for fitness-plan generation.

• Activity 1.3: Define the architecture connecting the frontend, backend, and Gemini AI.

• Activity 1.4: Set up the development environment and required dependencies.

Milestone 2: Core Functionalities Development

• Activity 2.1: Develop the fitness-plan generation functionality.

• Activity 2.2: Add user-input validation and backend processing.

• Activity 2.3: Structure AI responses so that workout plans are easy to read.

Milestone 3: App.py Development

• Activity 3.1: Develop the main Flask application logic and routes.

• Activity 3.2: Connect the backend to the Gemini API.

• Activity 3.3: Handle errors and validate AI-generated responses.

Milestone 4: Frontend Development

• Activity 4.1: Design a simple and responsive FitBuddy user interface.

• Activity 4.2: Create forms for fitness goals, preferences, level, and available time.

• Activity 4.3: Display the generated fitness plan dynamically.

Milestone 5: Testing and Deployment

• Activity 5.1: Test the application locally with different valid inputs.

• Activity 5.2: Verify that the generated plans are displayed correctly.

• Activity 5.3: Prepare the application for deployment and demonstrate the working project.

Milestone 6: Conclusion

• Review the completed FitBuddy application and document the results.

• Identify possible future improvements and extensions.

Team Member Work Assignment

Name	Assigned Work
Kaviya sree.T	Project Lead & Full-Stack Development: requirements analysis, Gemini model/API integration, Flask backend development, frontend coordination, testing, debugging, documentation, and final project integration.

Conclusion

FitBuddy-AI Fitness Plan Generator using Gemini Models is designed as an AI-assisted fitness planning application. By combining a user-friendly interface with Gemini-based generation, the project demonstrates how Generative AI can be used to organize personalized fitness planning. Future versions can add user accounts, plan history, progress tracking, richer activity recommendations, and additional personalization features.